# TEMPEST: Porewater Nutrients

2023-2025 Samples

2026-04-27

## Contents

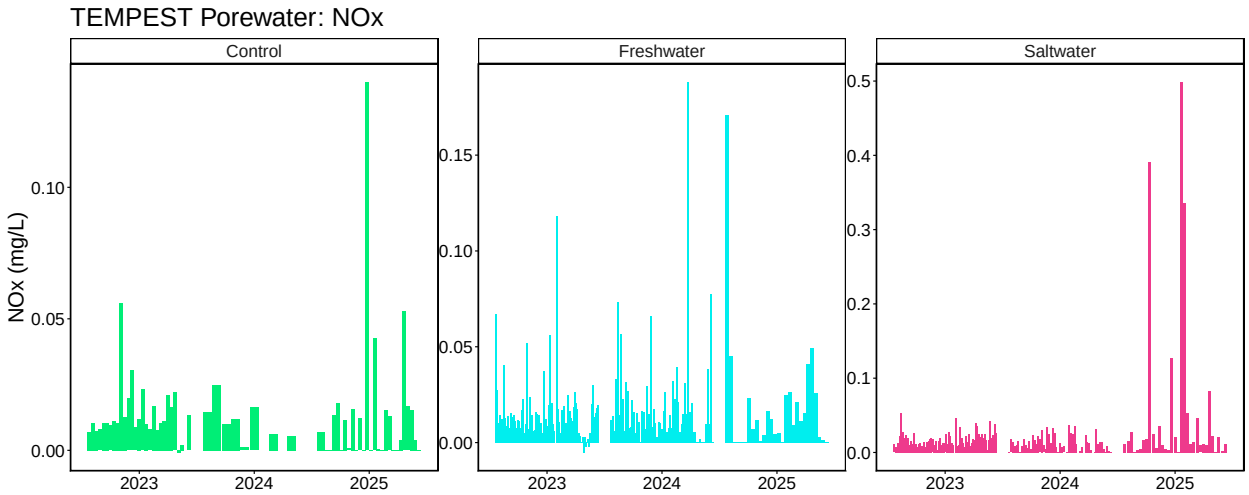

0.1 Grab 2022 and 2023 All Data Files . . . 1
0.2 Write out files . . . 2
0.3 Visualize Data by Plot . . . 2
0.4 Summarized data for Site and Zone . . . 4
0.5 Boxplot summary . . . 5
0.6 Boxplot summary for Sourcewater . . . 7
0.7 Calculate summary metrics . . . 8
0.8 Summary Tables for Porewater . . . 8
0.9 Summary Tables for Sourcewater . . . 8
0.10 Number of Samples . . . 9
0.11 Write out summarized data . . . 9

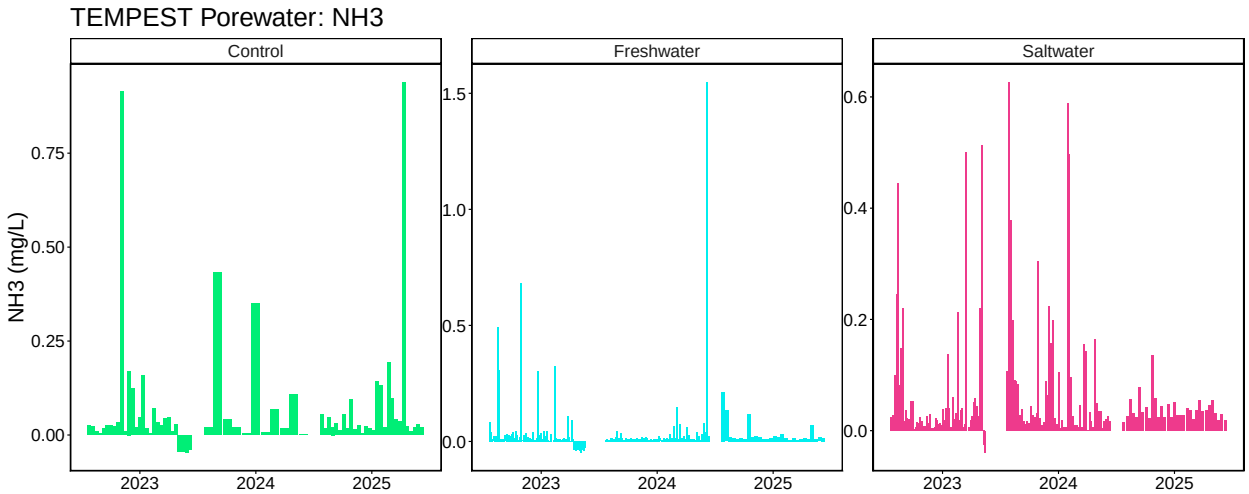

### 0.1 Grab 2022 and 2023 All Data Files

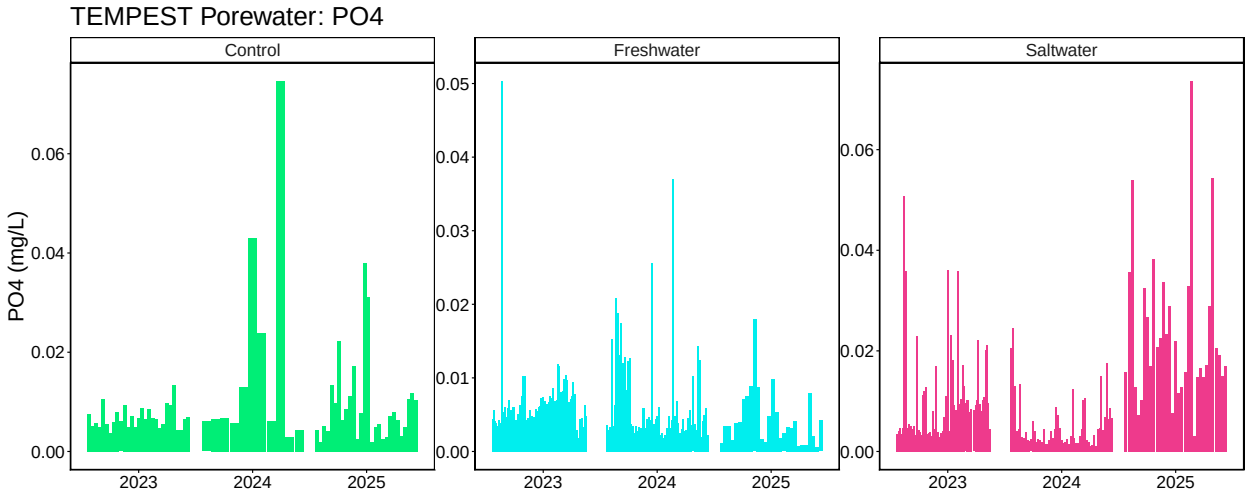
##Make Relevant Metadata Sheet

## 0.2 Write out files

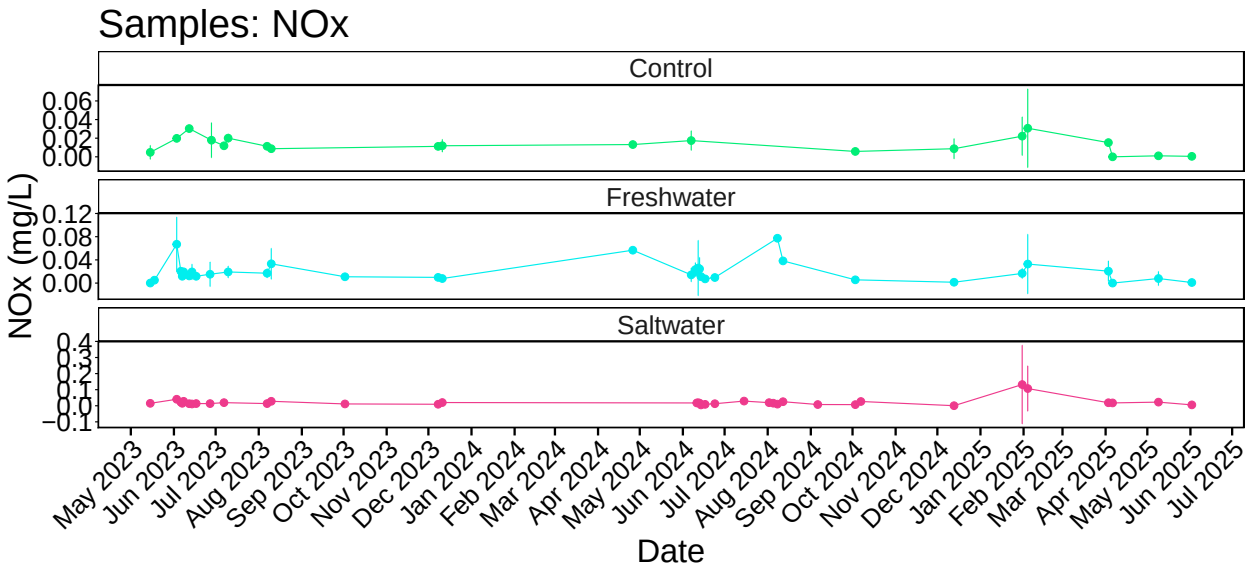
## 0.3 Visualize Data by Plot

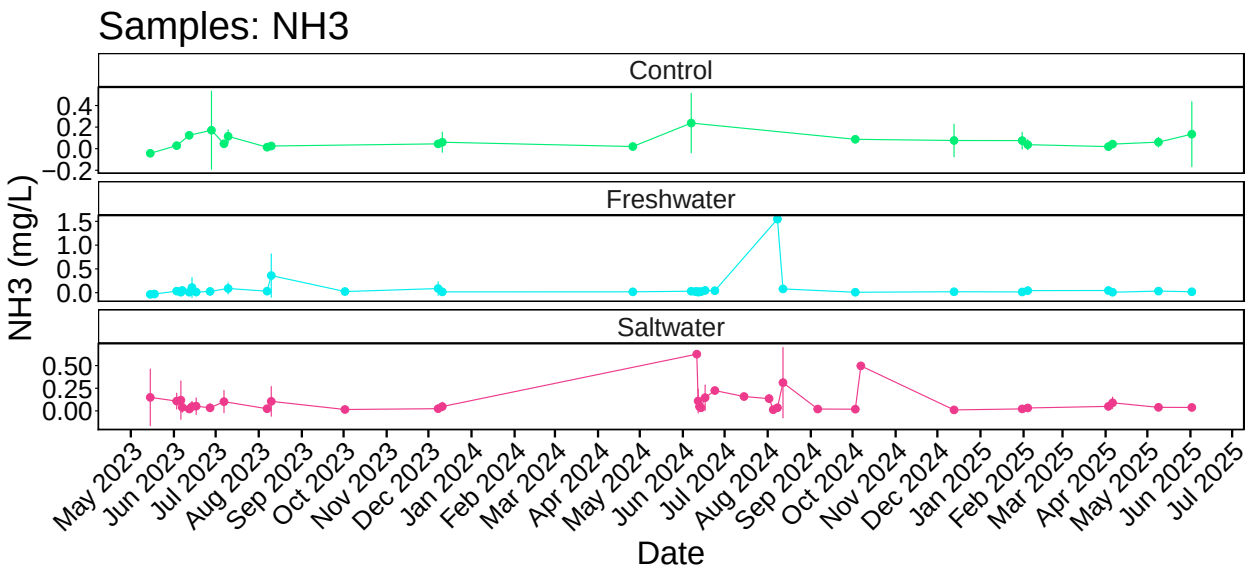


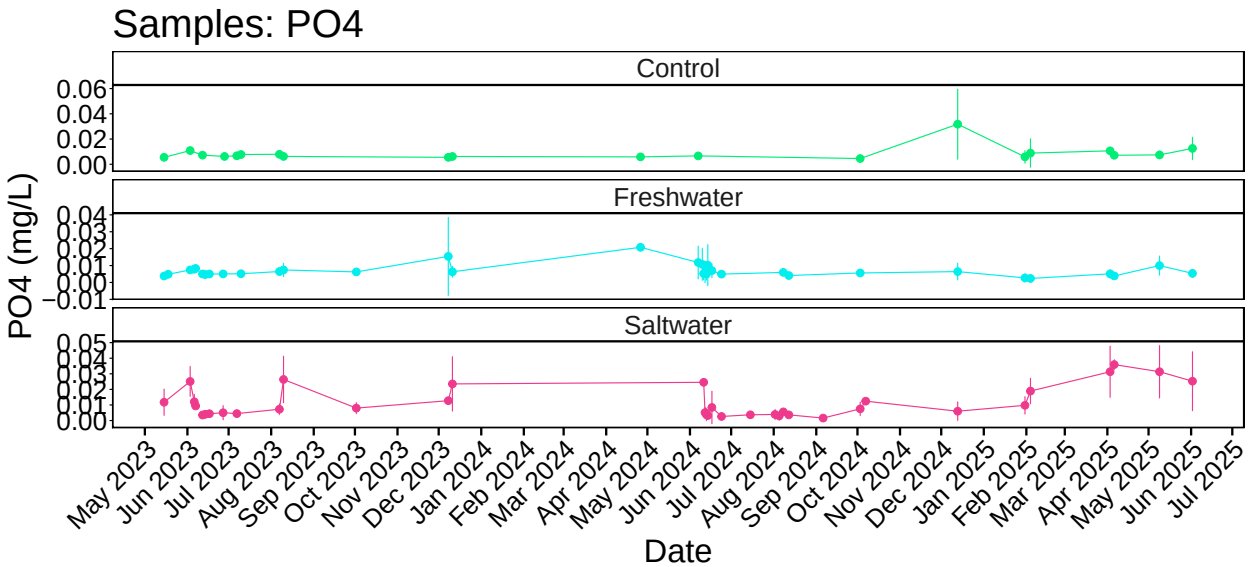



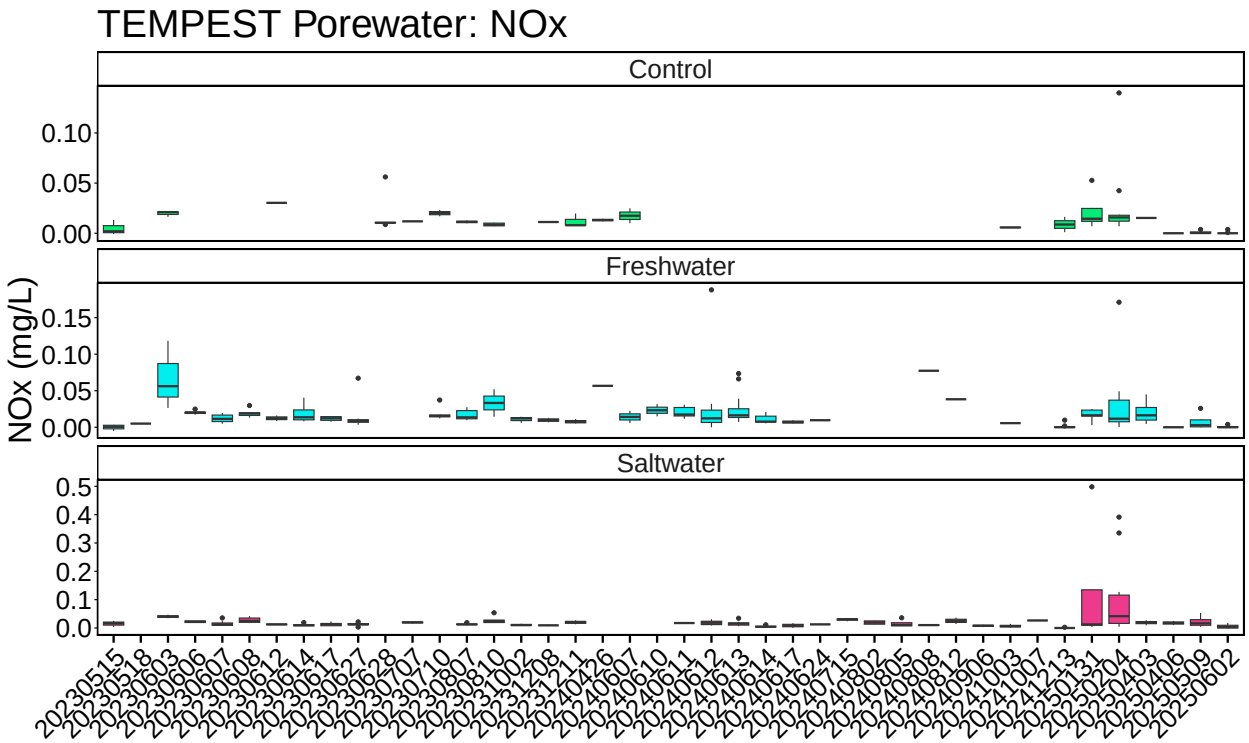
## 0.4 Summarized data for Site and Zone

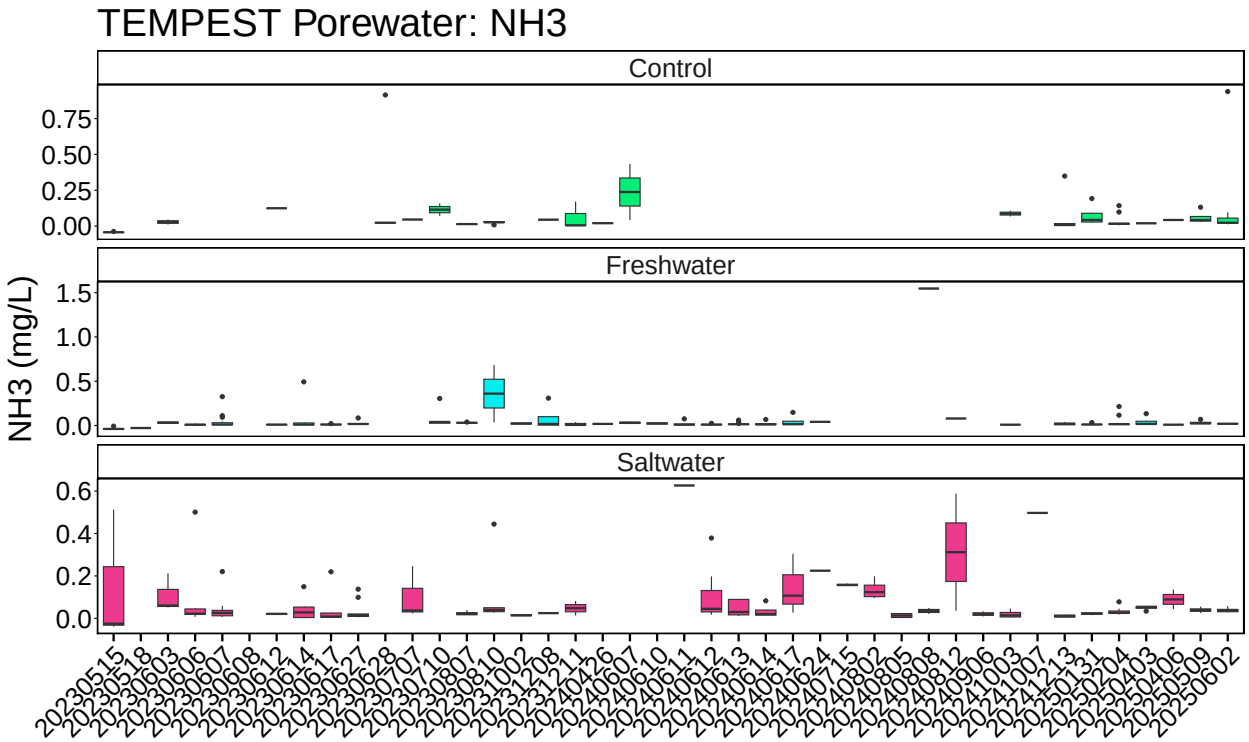


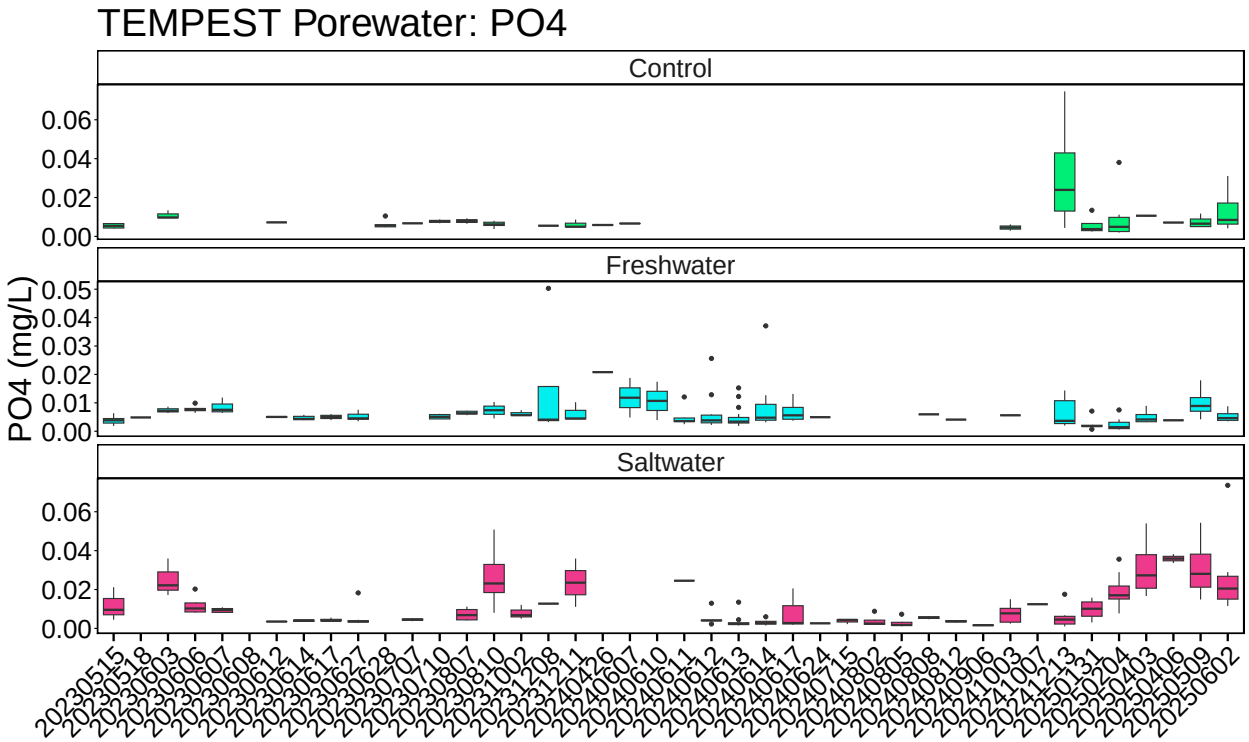



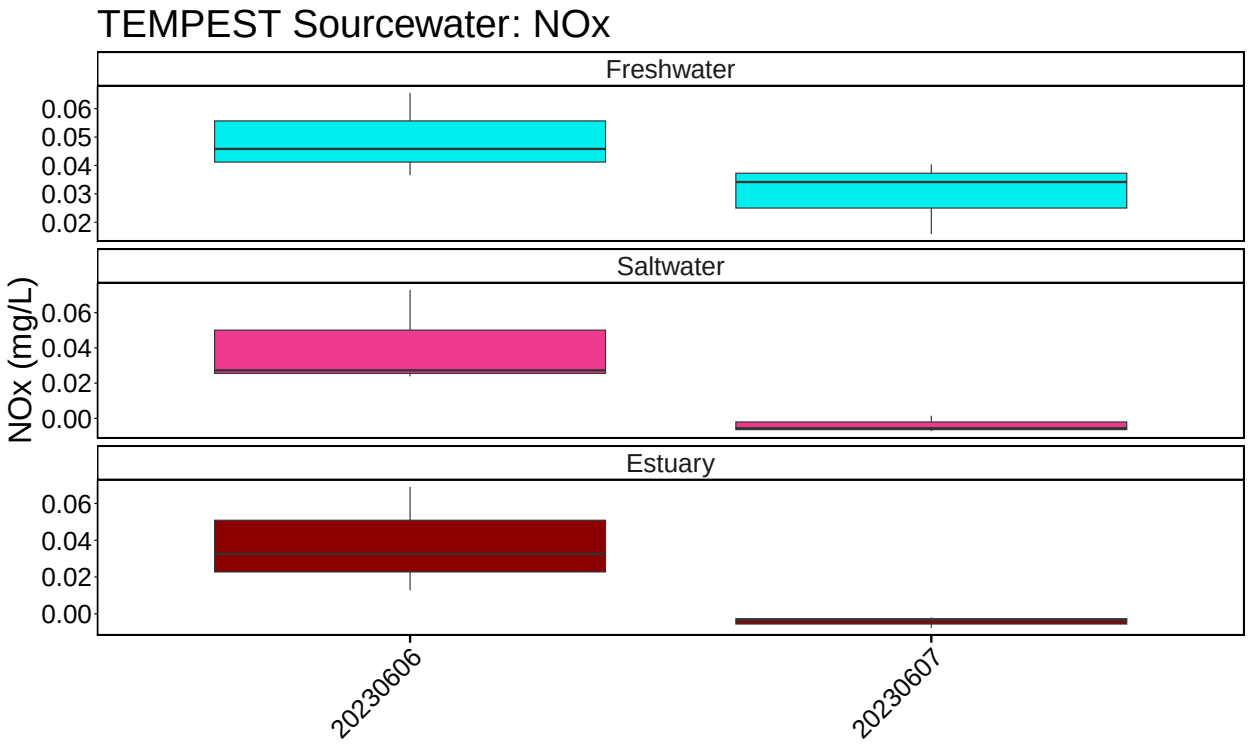
## 0.5 Boxplot summary

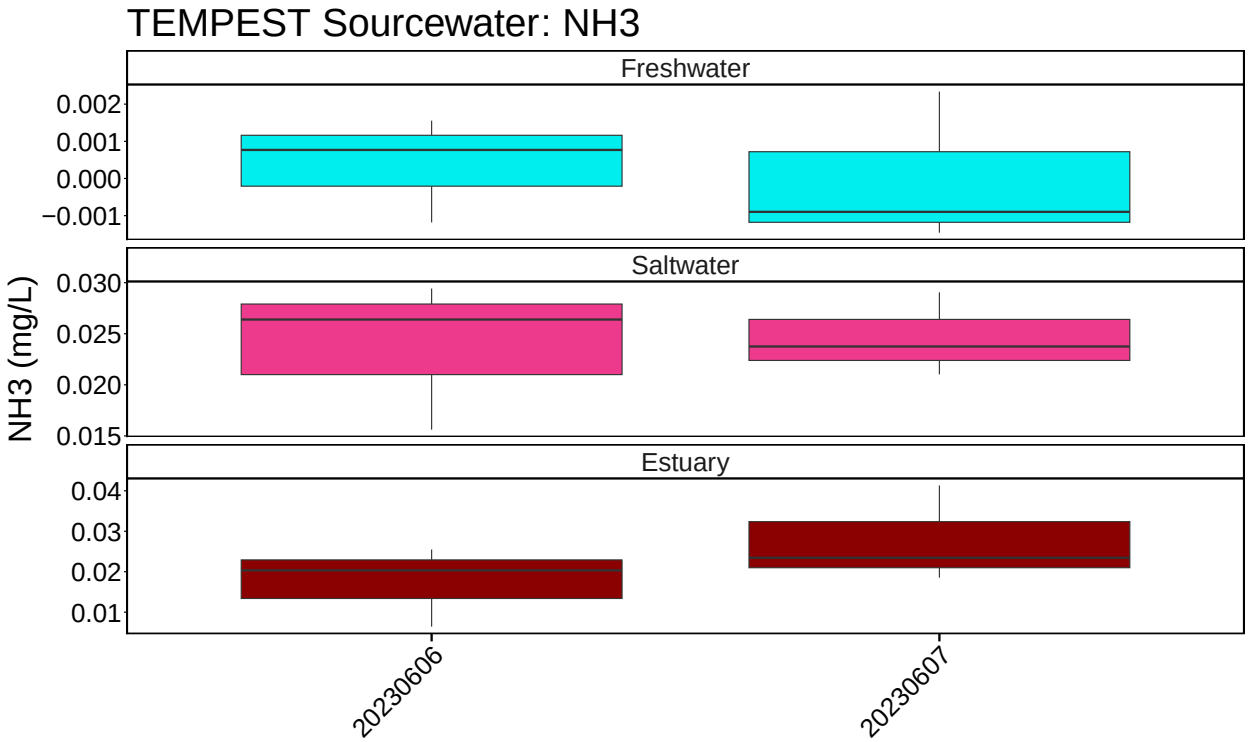


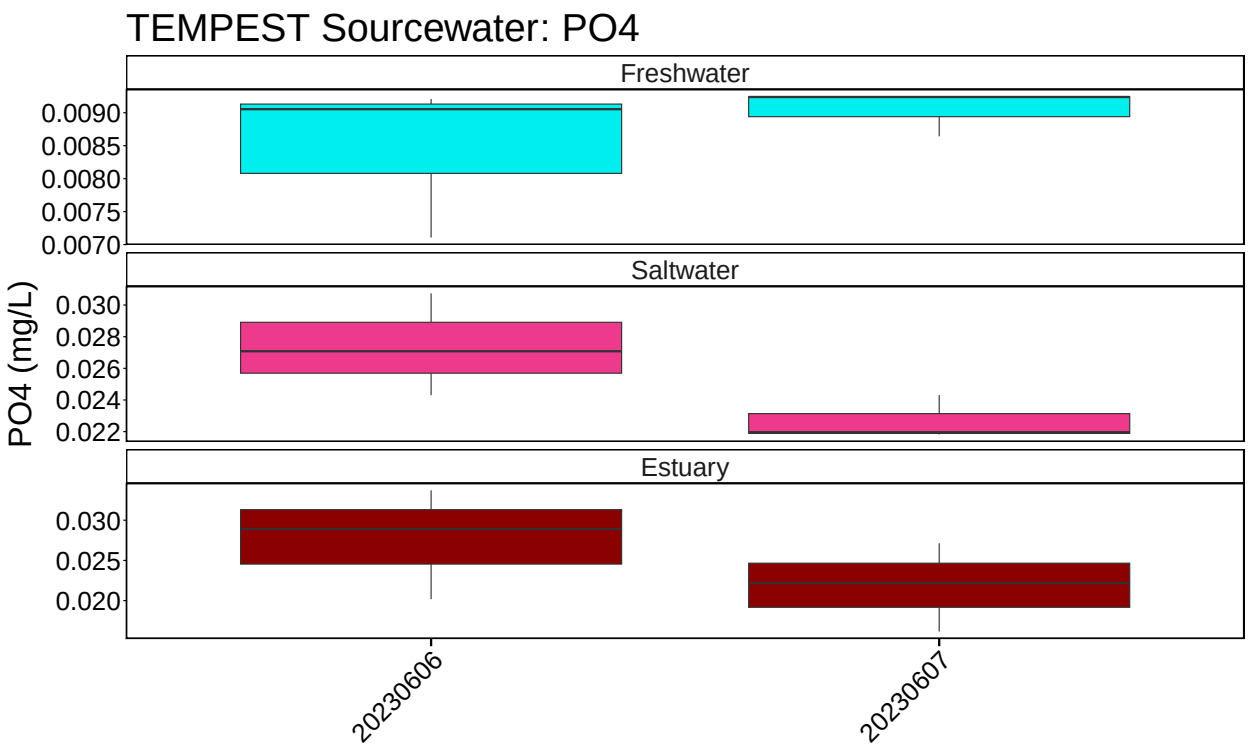




TEMPEST Porewater: PO4
Control
Freshwater
Saltwater
PO4 (mg/L)
0.00
0.02
0.04
0.06
0.01
0.03
0.05
20230515
20230518
20230603
20230606
20230607
20230608
20230612
20230614
20230617
20230627
20230628
20230707
20230710
20230807
20230810
20231002
20231208
20231211
20240426
20240607
20240610
20240611
20240612
20240613
20240614
20240617
20240624
20240715
20240802
20240805
20240808
20240812
20240906
20241003
20241007
20241213
20250131
20250204
20250403
20250406
20250509
20250602

## 0.6 Boxplot summary for Sourcewater






## 0.7 Calculate summary metrics

##Calculate summary metrics for Sourcewater

## 0.8 Summary Tables for Porewater

Table 1: Porewater Summary Statistics by Year

| Year | plot_1 | min_NOx_mgL | max_NOx_mgL | mean_NOx_mgL | min_NH3_mgL | max_NH3_mgL | mean_NH3_mgL | min_PO4_mgL | max_PO4_mgL | mean_PO4_mgL |
|---|---|---|---|---|---|---|---|---|---|---|
| **2023** | Control | -0.000815 | 0.056143 | 0.0137446 | -0.046479 | 0.914771 | 0.0600114 | 0.003797 | 0.013431 | 0.0068410 |
| **2023** | Freshwater | -0.005152 | 0.118250 | 0.0156258 | -0.047481 | 0.684655 | 0.0427431 | 0.001857 | 0.050299 | 0.0066750 |
| **2023** | Saltwater | 0.003158 | 0.053738 | 0.0181549 | -0.039194 | 0.512489 | 0.0608517 | 0.002830 | 0.050842 | 0.0104014 |
| **2024** | Control | 0.001067 | 0.024870 | 0.0112690 | 0.000633 | 0.433686 | 0.0971469 | 0.002947 | 0.074597 | 0.0175461 |
| **2024** | Freshwater | 0.000000 | 0.187981 | 0.0196821 | 0.005095 | 1.547253 | 0.0453378 | 0.001872 | 0.037088 | 0.0067932 |
| **2024** | Saltwater | 0.000000 | 0.036574 | 0.0135412 | 0.000458 | 0.626225 | 0.0893877 | 0.000997 | 0.024531 | 0.0052899 |
| **2025** | Control | 0.000000 | 0.139916 | 0.0138901 | 0.005014 | 0.938518 | 0.0767885 | 0.001983 | 0.038051 | 0.0094232 |
| **2025** | Freshwater | 0.000000 | 0.171113 | 0.0189085 | 0.005980 | 0.214552 | 0.0323748 | 0.000612 | 0.017934 | 0.0043941 |
| **2025** | Saltwater | 0.000000 | 0.498290 | 0.0560104 | 0.014578 | 0.135581 | 0.0390355 | 0.003052 | 0.073587 | 0.0235702 |

Table 2: Porewater Summary Statistics by Plot

| plot_1 | min_NOx_mgL | max_NOx_mgL | mean_NOx_mgL | min_NH3_mgL | max_NH3_mgL | mean_NH3_mgL | min_PO4_mgL | max_PO4_mgL | mean_PO4_mgL |
|---|---|---|---|---|---|---|---|---|---|
| **Control** | -0.000815 | 0.139916 | 0.0134988 | -0.046479 | 0.938518 | 0.0729268 | 0.001983 | 0.074597 | 0.0096359 |
| **Freshwater** | -0.005152 | 0.187981 | 0.0176697 | -0.047481 | 1.547253 | 0.0419696 | 0.000612 | 0.050299 | 0.0063310 |
| **Saltwater** | 0.000000 | 0.498290 | 0.0245543 | -0.039194 | 0.626225 | 0.0667009 | 0.000997 | 0.073587 | 0.0113253 |

## 0.9 Summary Tables for Sourcewater

##Calculate number of samples

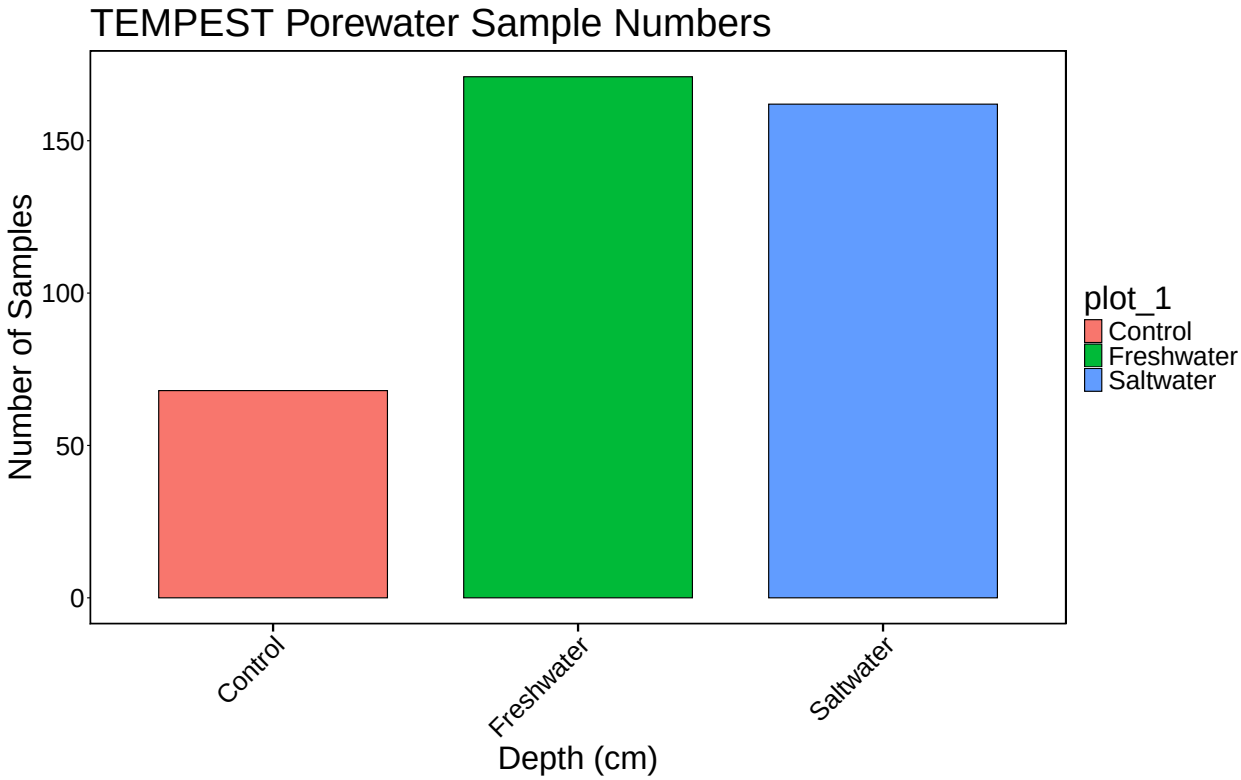
Table 3: Sourcewater Summary Statistics by Year

| Year | plot_1 | min_NOx_mgL | max_NOx_mgL | mean_NOx_mgL | min_NH3_mgL | max_NH3_mgL | mean_NH3_mgL | min_PO4_mgL | max_PO4_mgL | mean_PO4_mgL |
|---|---|---|---|---|---|---|---|---|---|---|
| **2023** | Freshwater | 0.0158 | 0.0655 | 0.0397 | -0.0015 | 0.0023 | 0.0002 | 0.0071 | 0.0092 | 0.0087 |
| **2023** | Saltwater | -0.0072 | 0.0729 | 0.0188 | 0.0156 | 0.0294 | 0.0242 | 0.0218 | 0.0307 | 0.0250 |
| **2023** | Estuary | -0.0077 | 0.0690 | 0.0169 | 0.0064 | 0.0413 | 0.0226 | 0.0162 | 0.0337 | 0.0247 |

## 0.10 Number of Samples

Table 4: Total Samples: 423

| Year | Number_of_samples |
|---|---|
| **2023** | 181 |
| **2024** | 131 |
| **2025** | 89 |




## 0.11 Write out summarized data